

Assignment No. 2

2.1-2.2

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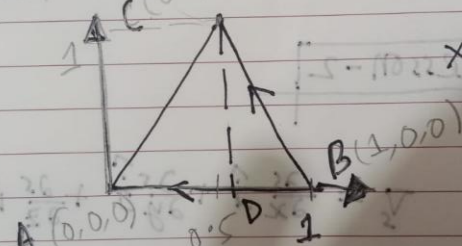
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solution

- 2.1(a)

volume = $\frac{1}{2} \times 1 \times 0.5 = 0.25$

②



xy plane
(no z component)

$\Delta = \frac{1}{2} \times AB \times CD = \frac{1}{2} \times AB \times BC \sin \theta$

$= \frac{1}{2} \times |\vec{AB} \times \vec{BC}|$

$\vec{AB} = \hat{i}$; $\vec{BC} = -0.5\hat{i} + \hat{j}$

$\Delta = \frac{1}{2} \|\vec{AB} \times \vec{BC}\| = \frac{1}{2} \text{ units}$

solution

- 2.1(b)

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(b)

$A(0,0,1)$
 $B(1,0,0)$
 $C(0.5,1,0)$
 $D(0.5,0.5,1)$

$-\vec{AB} = -\hat{i} \Rightarrow \vec{BA} = -\hat{i}$
 $\vec{BC} = -0.5\hat{i} + \hat{j}$
 $\vec{BD} = -0.5\hat{i} + 0.5\hat{j} + \hat{k}$

$\vec{BA} \cdot (\vec{BC} \times \vec{BD}) = \begin{vmatrix} -1 & 0 & 0 \\ -0.5 & 1 & 0 \\ -0.5 & 0.5 & 1 \end{vmatrix}$

$= 1 \text{ cubic unit}$
 $= \text{parallelepiped volume}$

pyramid volume $= \frac{1}{2} \times \frac{1}{3} \times 1$
 $= \frac{1}{6} \text{ cubic units}$

solution

- 2.2

